

HW#2: The Probability that a Matrix is Invertible

Note. Recall that

$$|\text{Mat}_n(\mathbb{F}_q)| = q^{n^2}, \quad |GL_n(\mathbb{F}_q)| = \prod_{i=0}^{n-1} (q^n - q^i),$$

and hence

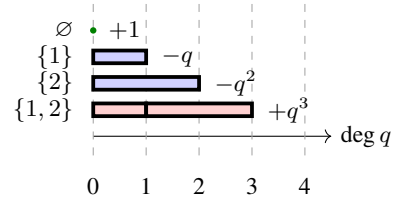
$$\begin{aligned} \frac{|GL_n(\mathbb{F}_q)|}{|\text{Mat}_n(\mathbb{F}_q)|} &= \frac{1}{q^{n^2}} \left[(q^n - 1)(q^n - q) \cdots (q^n - q^{n-1}) \right] = \frac{1}{q^{n^2}} \left[q^n \left(1 - \frac{1}{q^n}\right) \left(1 - \frac{1}{q^{n-1}}\right) \cdots \left(1 - \frac{1}{q^1}\right) \right] \\ &= \left(1 - \frac{1}{q^n}\right) \left(1 - \frac{1}{q^{n-1}}\right) \cdots \left(1 - \frac{1}{q^1}\right) \\ &= \prod_{j=1}^n (1 - q^{-j}). \end{aligned}$$

Observation. $(1 - q)(1 - q^2) = 1 - q - q^2 + q^3$. Let

$$\mathcal{P}_2 := 2^{\{1,2\}} = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}.$$

Then

$$\begin{aligned} 1 - q - q^2 + q^3 &= (-1)^0 q^0 + (-1)^1 q^1 + (-1)^1 q^2 + (-1)^2 q^3 \\ &= (-1)^{\#\emptyset} q^0 + (-1)^{\#\{1\}} q^1 + (-1)^{\#\{2\}} q^2 + (-1)^{\#\{1,2\}} q^{1+2} \\ &= \sum_{S \in \mathcal{P}_2} (-1)^{\#S} q^{\sum_{j \in S} j}. \end{aligned}$$

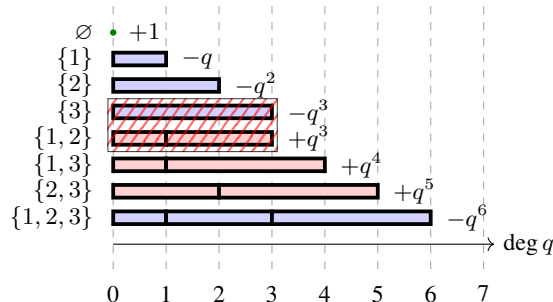


Observation. $(1 - q)(1 - q^2)(1 - q^3) = 1 - q - q^2 + q^4 + q^5 - q^6$. Let

$$\mathcal{P}_3 := 2^{\{1,2,3\}} = \{\emptyset, \{1\}, \{2\}, \{3\}, \dots, \{1,2,3\}\}.$$

Then

$$\begin{aligned} 1 - q - q^2 + q^4 + q^5 - q^6 &= (-1)^0 q^0 + (-1)^1 q^1 + (-1)^1 q^2 + (-1)^1 q^3 \\ &\quad + (-1)^2 q^3 + (-1)^2 q^4 + (-1)^2 q^5 + (-1)^3 q^6 \\ &= (-1)^{\#\emptyset} q^0 + (-1)^{\#\{1\}} q^1 + (-1)^{\#\{2\}} q^2 + (-1)^{\#\{3\}} q^3 \\ &\quad + (-1)^{\#\{1,2\}} q^{1+2} + (-1)^{\#\{1,3\}} q^{1+3} + (-1)^{\#\{2,3\}} q^{2+3} + (-1)^{\#\{1,2,3\}} q^{1+2+3} \\ &= \sum_{S \in \mathcal{P}_3} (-1)^{\#S} q^{\sum_{j \in S} j}. \end{aligned}$$



Observation. $(1-q)(1-q^2)(1-q^3)(1-q^4) = 1 - q - q^2 + 2q^5 - q^8 - q^9 + q^{10}$. Let

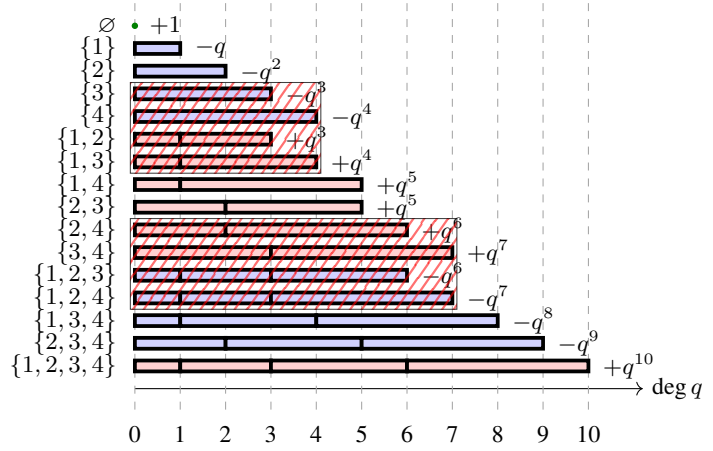
$$\mathcal{P}_4 := 2^{\{1,2,3,4\}} = \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \dots, \{1,2,3,4\}\}.$$

Then

$$\begin{aligned} \prod_{i=1}^4 (1-q^i) &= (1-q)(1-q^2)(1-q^3)(1-q^4) \\ &= \sum_{S \in \mathcal{P}_4} (-1)^{\#S} q^{\sum_{j \in S} j}. \end{aligned}$$

Thus, we have

$$\begin{aligned} \prod_{i=1}^N (1-q^i) &= \sum_{S \in \mathcal{P}_N} (-1)^{\#S} q^{\sum_{j \in S} j} \\ &= \sum_{S \subseteq \{1,2,\dots,N\}} (-1)^{\#S} q^{\sum_{j \in S} j}. \end{aligned}$$



Pentagonal Number Theorem

Lemma 1.

$$\prod_{j=1}^{\infty} (1-q^j) = \sum_{k=-\infty}^{\infty} (-1)^k q^{k(3k-1)/2}.$$

Proof. Recall that

$$\prod_{j=1}^N (1-q^j) = \sum_{S \in \mathcal{P}_N} (-1)^{\#S} q^{\sum_{j \in S} j}.$$

For each $n \geq 0$, let

$$x_n := \sum_{\substack{S \subseteq \{1,\dots,N\} \\ \sum_{j \in S} j = n}} (-1)^{\#S}$$

be the coefficient of q^n . We want to compute

$$\begin{aligned} \prod_{j=1}^{\infty} (1-q^j) &= \lim_{N \rightarrow \infty} \sum_{S \in \mathcal{P}_N} (-1)^{\#S} q^{\sum_{j \in S} j} \\ &= \sum_{n=0}^{\infty} x_n q^n, \quad \text{where } x_n := \sum_{\substack{S \subseteq \mathbb{N} \text{ with } |S| < \infty \\ \sum_{j \in S} j = n}} (-1)^{\#S}. \end{aligned}$$

Let $S = \{a_1 < a_2 < \dots < a_r\} \subseteq \mathbb{N}$ be a finite nonempty subset. We define

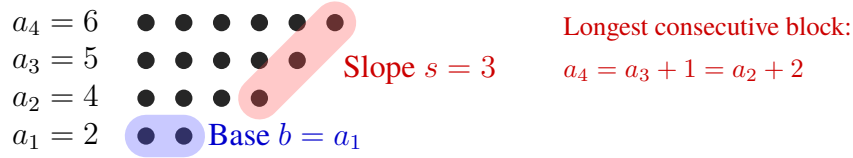
- the **base** of S by $b := a_1$, the smallest element of S ; and
- the **slope** of S to be the length s of the longest consecutive block at the top of S , that is,

$$s := \max\{t \geq 1 : a_r, a_{r-1}, \dots, a_{r-t+1} \text{ are consecutive integers}\}.$$

Equivalently,

$$a_r = a_{r-1} + 1 = \dots = a_{r-(s-1)} + (s-1).$$

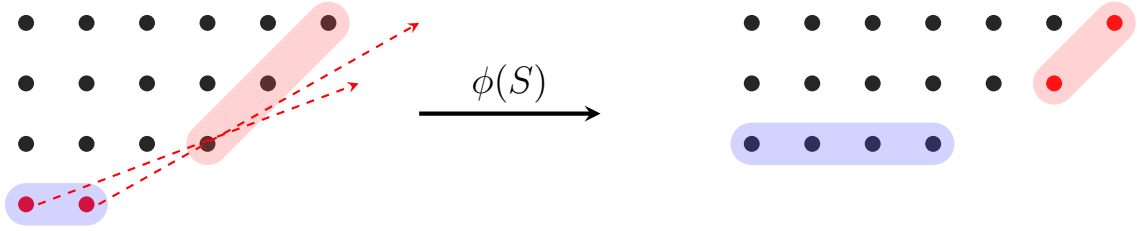
For example, if $S = \{a_1 < a_2 < a_3 < a_4\} = \{2, 4, 5, 6\}$, then



$b = 2$ and $s = 3$. We now define a map ϕ .

Case 1: $b < s$. Remove the smallest element b , and add 1 to each of the b largest elements.

For example, if $S = \{a_1 < a_2 < a_3 < a_4\} = \{2, 4, 5, 6\}$, then $b = 2$ and $s = 3$.



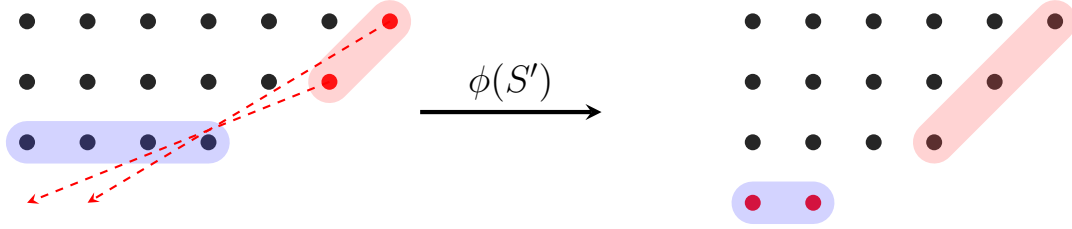
Weight: $(-1)^{\#S} q^{\sum j} = +q^{17}$

Weight: $(-1)^{\#\phi(S)} q^{\sum j} = -q^{17}$

Then $\phi(S) = \{4, 6, 7\}$

Case 2: $b > s$. Subtract 1 from each of the s largest elements, and adjoin the element s .

For example, if $S' = \{a_1 < a_2 < a_3 < a_4\} = \{4, 5, 7\}$, then $b = 4$ and $s = 2$.



Weight: $(-1)^{\#S'} q^{\sum j} = -q^{17}$

Weight: $(-1)^{\#\phi(S')} q^{\sum j} = +q^{17}$

Then $\phi(S) = \{2, 4, 5, 6\}$.

By construction, $\sum_{j \in \phi(S)} j = \sum_{j \in S} j$. Moreover,

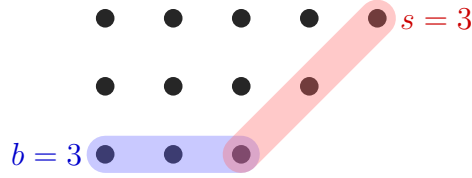
$$\#\phi(S) = \begin{cases} \#S - 1 & \text{if } b < s \\ \#S + 1 & \text{if } b > s \end{cases}$$

Hence

$$(-1)^{\#\phi(S)} q^{\sum_{j \in \phi(S)} j} = \begin{cases} (-1)^{-1} (-1)^{\#S} q^{\sum_{j \in \phi(S)} j} & \text{if } b < s \\ (-1)^{+1} (-1)^{\#S} q^{\sum_{j \in \phi(S)} j} & \text{if } b > s \end{cases} = -(-1)^{\#S} q^{\sum_{j \in S} j}.$$

The only subsets for which ϕ is not defined are the exceptional ones:

- (i) ($b = s$) Let $S = \{k, k+1, \dots, 2k-1\}$ ($k \geq 1$). For example, let $S = \{3, 4, 5\}$.



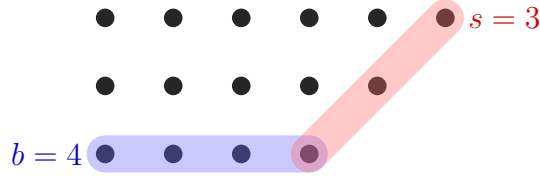
The Base and Slope intersect at the exact same element. Since

$$k + (k+1) + \dots + (2k-1) = \frac{k}{2} \cdot (3k-1), \quad \text{and} \quad \#\{k, k+1, \dots, \underbrace{2k-1}_{=k+(k-1)}\} = k,$$

we have

$$\boxed{(-1)^{\#S} q^{\sum_{j \in S} j} = (-1)^k q^{k(3k-1)/2}}.$$

- (ii) ($b = s+1$) Let $S = \{k+1, k+2, \dots, 2k\}$ ($k \geq 1$). For example, let $S = \{4, 5, 6\}$.



The Base and Slope intersect. Since

$$(k+1) + (k+2) + \dots + 2k = \frac{k}{2} \cdot (3k+1), \quad \text{and} \quad \#\{k+1, k+2, \dots, \underbrace{2k}_{=k+k}\} = k,$$

we obtain

$$\boxed{(-1)^{\#S} q^{\sum_{j \in S} j} = (-1)^k q^{k(3k+1)/2}}.$$

- (iii) ($S = \emptyset$) The empty set has no elements, so it has neither a base b nor a slope s . Then

$$\boxed{(-1)^{\#\emptyset} q^{\sum_{j \in \emptyset} j} = (-1)^0 q^0 = 1}.$$

Hence

$$\begin{aligned} \prod_{j=1}^{\infty} (1 - q^j) &= \lim_{N \rightarrow \infty} \prod_{j=1}^N (1 - q^j) = \lim_{N \rightarrow \infty} \sum_{S \in \mathcal{P}_N} (-1)^{\#S} q^{\sum_{j \in S} j} \\ &= 1 + \sum_{k=1}^{\infty} (-1)^k q^{k(3k-1)/2} + \sum_{k=1}^{\infty} (-1)^k q^{k(3k+1)/2} \\ &= \underbrace{(-1)^0 q^0}_{k=0} + \sum_{k=1}^{\infty} (-1)^k q^{k(3k-1)/2} + \sum_{k=-\infty}^{-1} (-1)^{-k} q^{-k(3(-k)+1)/2} \\ &= \sum_{k=-\infty}^{-1} (-1)^k q^{k(3k-1)/2} + (-1)^0 q^0 + \sum_{k=1}^{\infty} (-1)^k q^{k(3k-1)/2} \\ &= \sum_{k=-\infty}^{\infty} (-1)^k q^{k(3k-1)/2}. \end{aligned}$$

□

Pentagonal Number Theorem

Theorem 2. *The series*

$$\begin{aligned}\prod_{j=1}^{\infty} (1 - x^j) &= 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{k(3k-1)/2} + x^{k(3k+1)/2} \right) \\ &= \sum_{k=-\infty}^{\infty} (-1)^k x^{k(3k-1)/2}.\end{aligned}$$

converge for $|x| < 1$ *Proof.* Note that

$$\sum |a_n| < \infty \implies \sum a_n \text{ converges.}$$

We want to prove that the series

$$1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{k(3k-1)/2} + x^{k(3k+1)/2} \right) \text{ converges absolutely when } |x| < 1.$$

Indeed,

$$\sum_{k=1}^{\infty} \left| x^{k(3k-1)/2} + x^{k(3k+1)/2} \right| \leq \sum_{k=1}^{\infty} \left(|x|^{k(3k-1)/2} + |x|^{k(3k+1)/2} \right).$$

Since for every $k \geq 1$,

$$\frac{k(3k-1)}{2} \geq k, \quad \frac{k(3k+1)}{2} \geq k \implies |x|^{k(3k-1)/2} \leq |x|^k, \quad |x|^{k(3k+1)/2} \leq |x|^k.$$

Therefore

$$\sum_{k=1}^{\infty} \left(|x|^{k(3k-1)/2} + |x|^{k(3k+1)/2} \right) \leq 2 \sum_{k=1}^{\infty} |x|^k.$$

The geometric series on the right converges because $|x| < 1$. Hence

$$1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{k(3k-1)/2} + x^{k(3k+1)/2} \right) \text{ converges absolutely for } |x| < 1.$$

□

Pentagonal Number Theorem

Theorem 3. *Let $n \geq 2$. Then*

$$\Pr[\mathbf{M} \stackrel{\$}{\leftarrow} \text{Mat}_n(\mathbb{F}_2) : \det(\mathbf{M}) \neq 0] > 0.288788.$$

Proof. Define

$$P_n := \prod_{j=1}^n (1 - 2^{-j})$$

Since P_n is a decreasing sequence, we have

$$P_n > \prod_{j=1}^{\infty} (1 - 2^{-j}) \text{ for every } n \geq 2.$$

Applying pentagonal-number theorem, we obtain

$$\prod_{j=1}^{\infty} (1 - 2^{-j}) = 1 + \sum_{k=1}^{\infty} (-1)^k \left(2^{-k(3k-1)/2} + 2^{-k(3k+1)/2} \right).$$

Therefore

$$\begin{aligned} \prod_{j=1}^{\infty} (1 - 2^{-j}) &= 1 - (2^{-1} + 2^{-2}) + (2^{-5} + 2^{-7}) - (2^{-12} + 2^{-15}) + (2^{-22} + 2^{-26}) - \dots \\ &> 1 - (2^{-1} + 2^{-2}) + (2^{-5} + 2^{-7}) - (2^{-12} + 2^{-15}) + 2^{-22} \\ &= 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{32} + \frac{1}{128} - \frac{1}{4096} - \frac{1}{32768} + \frac{1}{4194304} \\ &= 0.2887880802154541\dots \\ &> 0.288788. \end{aligned}$$

Since $P_n > \prod_{j=1}^{\infty} (1 - 2^{-j})$, it follows that

$$\Pr[\mathbf{M} \xleftarrow{\$} \text{Mat}_n(\mathbb{F}_2) : \det(\mathbf{M}) \neq 0] = \prod_{j=1}^n (1 - 2^{-j}) > 0.288788.$$

□

```
def pentagonal_series(N):
    R = RealField(50)
    return R(1) + sum(
        (-1)^k * (R(2)^(-k*(3*k-1)/2)
        + R(2)^(-k*(3*k+1)/2))
        for k in range(1, N+1)
    )

for r in range(1, 11):
    s = pentagonal_series(r)
    print(f"(1 to {r+1:2d}) partial sum = {s}")
```

```
(1 to 2) partial sum = 0.250000000000000
(1 to 3) partial sum = 0.289062500000000
(1 to 4) partial sum = 0.28878784179688
(1 to 5) partial sum = 0.28878809511662
(1 to 6) partial sum = 0.28878809508660
(1 to 7) partial sum = 0.28878809508660
(1 to 8) partial sum = 0.28878809508660
(1 to 9) partial sum = 0.28878809508660
(1 to 10) partial sum = 0.28878809508660
(1 to 11) partial sum = 0.28878809508660
```

```
def bilateral_pentagonal_series(N):
    R = RealField(50)
    return sum(
        (-1)^k * R(2)^(-k*(3*k-1)/2)
        for k in range(-N, N+1)
    )

for r in range(1, 11):
    s = bilateral_pentagonal_series(r)
    print(f"(-{r:2d} to {r+1:2d}) partial sum =
    ↪ {s}")
```

```
(- 1 to 2) partial sum = 0.250000000000000
(- 2 to 3) partial sum = 0.289062500000000
(- 3 to 4) partial sum = 0.28878784179688
(- 4 to 5) partial sum = 0.28878809511662
(- 5 to 6) partial sum = 0.28878809508660
(- 6 to 7) partial sum = 0.28878809508660
(- 7 to 8) partial sum = 0.28878809508660
(- 8 to 9) partial sum = 0.28878809508660
(- 9 to 10) partial sum = 0.28878809508660
(-10 to 11) partial sum = 0.28878809508660
```